CYCLISTIC BIKE-SHARE

Rider Behavior & Trend (2026)

How Do Annual Members and Casual Riders Use Cyclistic Bikes Differently?

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1. Executive Summary

Cyclistic's marketing director believes that future growth depends on increasing the number of annual memberships, because finance analysts have found members to be far more profitable than casual riders. This report analyzes Cyclistic trip data to answer the first guiding question: how do annual members and casual riders use Cyclistic bikes differently?

The analysis covers 6.01 million rides, of which members account for 3.90 million (64%) and casual riders for 2.11 million (35%). Three patterns stand out. First, members ride mainly on weekdays, while casual ridership rises sharply on weekends and peaks on Saturday. Second, casual riders take longer rides in time, especially on weekends. Third, members show a commuter-style hourly pattern with peaks at 8 AM and 5 PM, whereas casual riders build up gradually through the day. Electric bikes make up about 70% of all rides and are popular with both groups.

Together, these patterns suggest that members use Cyclistic mainly as a routine, weekday transport option, while casual riders use it more for leisure. The recommendations in Section 7 build on this contrast.

2. Business Task

The business task is to identify how annual members and casual riders use Cyclistic bikes differently, so that the marketing team can design a strategy to convert casual riders into annual members. The key stakeholders are Lily Moreno (Director of Marketing), the Cyclistic marketing analytics team, and the Cyclistic executive team, who must approve any recommended program.

3. Data Sources

The analysis uses Cyclistic's historical bike trip data, a public dataset made available by Motivate International Inc. under its published license. The data is organized as one record per ride and includes the ride start and end times, bike type, and rider type (member or casual). Data-privacy rules prohibit the use of personally identifiable information, so rides cannot be linked to individual riders. As a result, it is not possible to tell whether casual riders live in the service area or have bought multiple single passes.

The data is credible and comprehensive in that it covers the whole network, but it has limits. It describes rides, not riders, so the same person may appear many times. Ride duration is a measure of time, not distance. These limits are considered when interpreting the findings below.

4. Data Cleaning and Manipulation

The following steps were used to prepare the data for analysis:

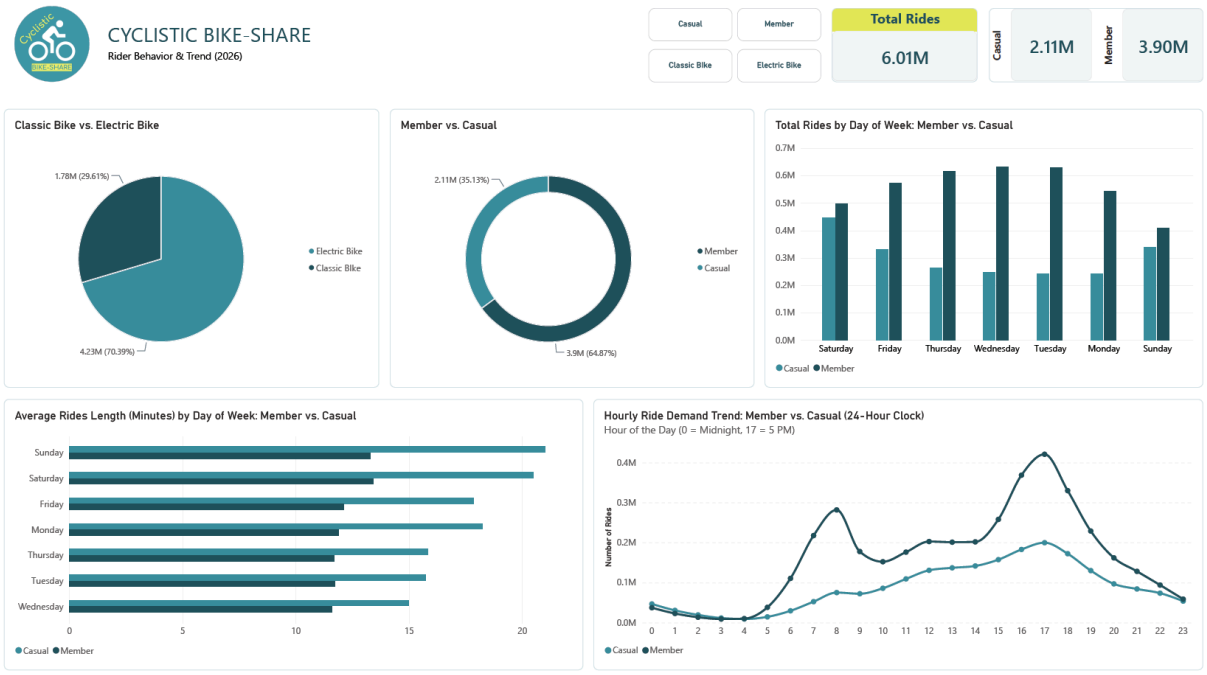
- Combined the monthly trip files into a single dataset covering the analysis period.
- Calculated ride length by subtracting the start time from the end time, and derived the day of the week and the hour of the day from the start time.
- Checked for and removed errors such as missing values, duplicate records, and rides with zero or negative duration.
- Aggregated rides by rider type, day of week, hour of day, and bike type.
- Used Python (pandas and seaborn) for analysis and charts, and Power BI to build the interactive dashboard.

5. Summary of Analysis

Rides were grouped by rider type and compared across four dimensions: day of the week, average ride length, hour of the day, and bike type. The table below summarizes the main differences.

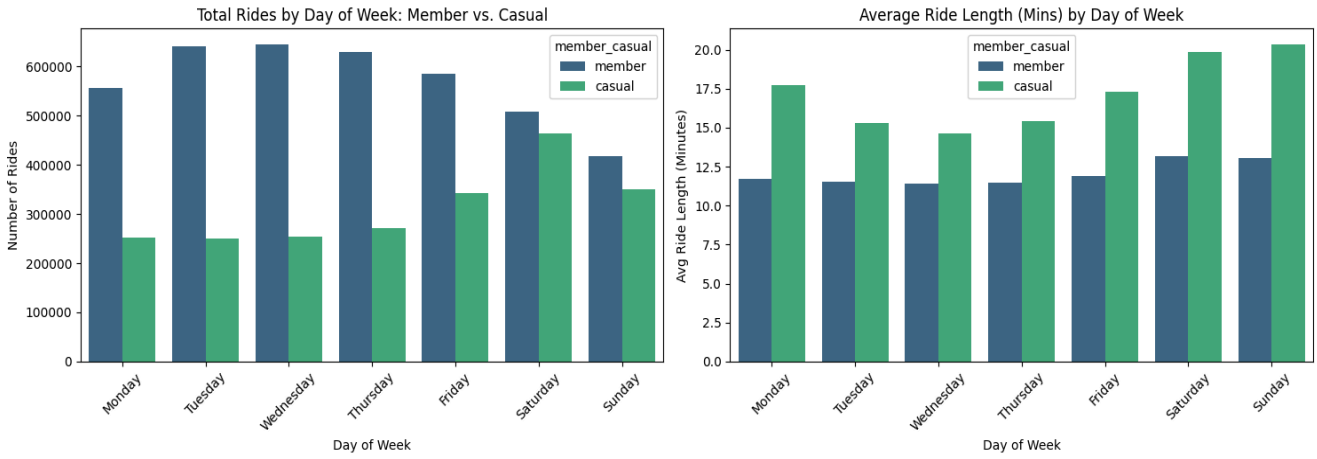
Measure	Casual riders	Annual members
Total rides	2.11M (35.13%)	3.90M (64.87%)
Average ride length	About 15–21 min	About 11.5–13 min
Weekly peak day	Saturday	Tuesday–Thursday
Hourly peaks	Gradual rise to 5 PM	8 AM and 5 PM

Figure 1 shows the Power BI dashboard that brings these views together, including the headline totals and the split between rider types and bike types.



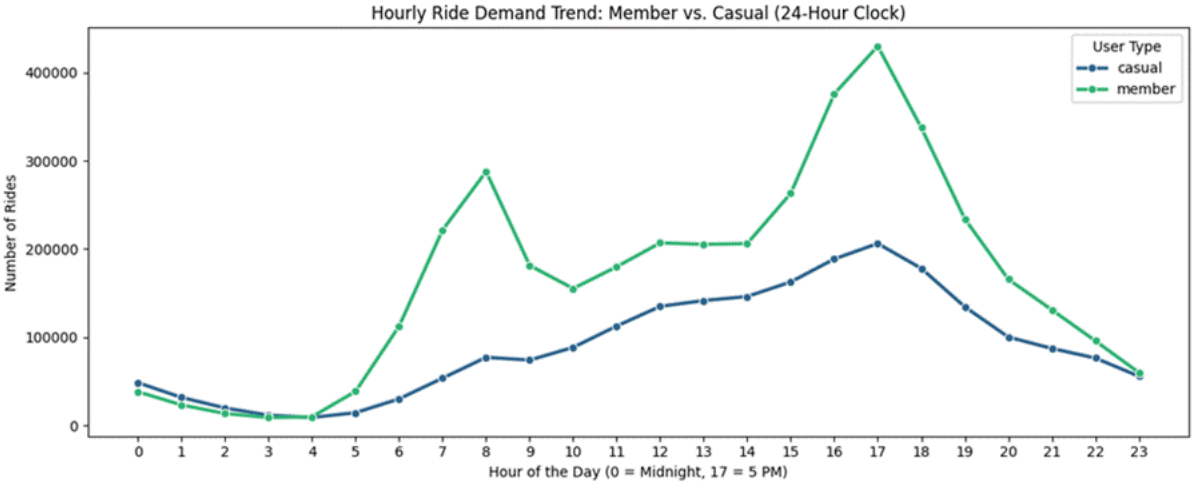
6. Visualizations and Key Findings

6.1 Rides by Day of Week and Average Ride Length



- **Members ride far more on weekdays.** From Monday to Thursday, members take roughly twice as many rides as casual riders. The gap narrows on Friday (about 1.7 times).
- **Casual riding peaks on the weekend.** Casual rides climb from about 0.25 million on a typical weekday to about 0.45 million on Saturday, nearly matching members. Sunday falls back to about 0.34 million.
- **Casual rides last longer.** Casual riders average roughly 15 to 18 minutes on weekdays and 20 to 21 minutes on weekends, compared with about 11.5 to 13 minutes for members on every day. Because this measures time, not distance, it shows that casual riders keep bikes longer, which is consistent with leisure riding, but it does not prove they travel farther.

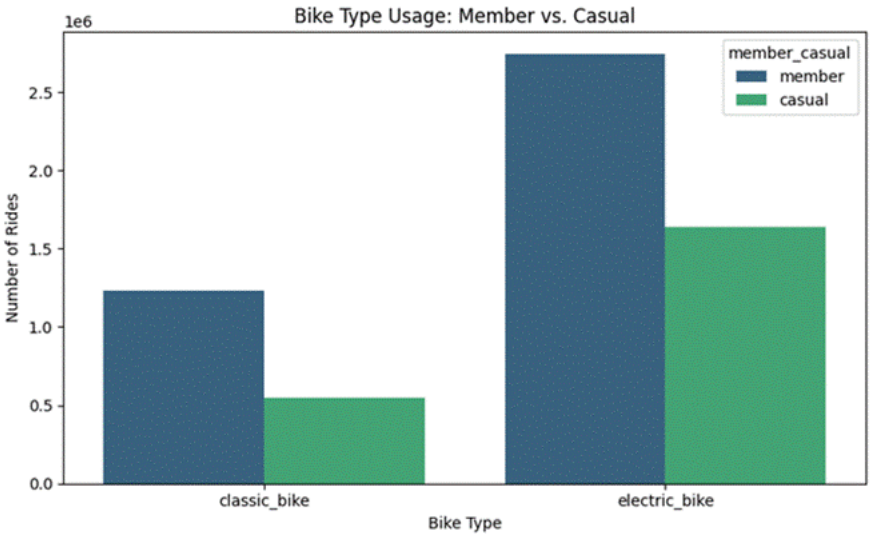
6.2 Hourly Demand



- **Members follow a commuter pattern.** Member rides rise from 5 AM to a morning peak at 8 AM (about 0.29 million), settle at a midday plateau of about 0.2 million, and then climb to a larger peak at 5 PM (about 0.42 million). Two peaks aligned with typical work and school hours point to commuting to and from work or school.
- **Casual riders build gradually.** Casual demand rises steadily from 5 AM to a single peak at 5 PM (about 0.2 million) and then declines gradually through the evening, with no sharp commuting spikes. This is consistent with leisure or exploratory riding, which has no fixed schedule.
- **Late night.** Casual riders slightly outnumber members between midnight and 3 AM, although volumes are very low for both groups.

These explanations are reasonable interpretations of the patterns, not proven facts. They could be tested further by looking at round trips, popular start stations, and separate weekday and weekend views.

6.3 Bike Type



- **Electric bikes dominate.** Electric bikes account for 4.23 million rides (70.39%) compared with 1.78 million (29.61%) for classic bikes. Members take more rides than casual riders on both bike types, and both groups prefer electric bikes by a similar margin.

Because the preference for e-bikes is shared by both groups, bike type does not separate members from casual riders. It is, however, a strong potential selling point for a membership offer.

7. Top Three Recommendations

1. **Launch a weekend-focused membership offer.** Casual ridership peaks on Saturday and remains high on Sunday. A weekend or flexible membership plan, promoted at the points where casual riders are most active, would meet them where they already ride.

2. **Show casual riders how much they could save.** Casual riders take longer rides, which are likely to cost more per trip under single-ride and day passes. In-app messages and emails sent after a long weekend ride can compare what the rider just paid with the cost of a membership.

3. **Promote e-bike benefits as a membership perk.** About 70% of all rides are on electric bikes, and casual riders choose them as often as members do. Offering member discounts or perks on e-bike rides gives casual riders a clear, concrete reason to join.

8. Limitations and Next Steps

- Ride length measures time, not distance. Calculating distance from start and end coordinates would show whether casual riders really travel farther.
- Average ride length can be skewed by a small number of very long rides. Reporting the median alongside the mean would make the comparison more robust.
- The hourly chart combines weekdays and weekends. Separate weekday and weekend views would likely sharpen the commuter pattern for members and reveal a midday leisure peak for casual riders on weekends.
- Monthly and seasonal trends were not analyzed. Chicago's weather probably affects casual riders strongly and should guide the timing of campaigns.
- Start-station data could show where casual riders begin their rides, which would help target marketing to the right locations.

9. Conclusion

Annual members use Cyclistic mainly for short, regular weekday trips that look like commuting, while casual riders take fewer but longer rides, concentrated on weekends and spread across the day. Both groups favor electric bikes. These differences point toward a marketing approach built around weekend leisure use, clear cost savings, and e-bike perks. Further analysis of seasonality, distance, and start stations would strengthen these recommendations before the executive team makes its decision.